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Appearance of chaos and hyperchaos in evolving pendulum network

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The study of deterministic chaos continues to be one of the important problems in the field of nonlinear dynamics. Interest in the study of chaos exists both in low-dimensional dynamical systems and in large ensembles of coupled oscillators. In this paper, we study the emergence of chaos in chains of locally coupled identical pendulums with constant torque. The study of the scenarios of the emergence (disappearance) and properties of chaos is done as a result of changes in: (i) the individual properties of elements due to the influence of dissipation in this problem, and (ii) the properties of the entire ensemble under consideration, determined by the number of interacting elements and the strength of the connection between them. It is shown that an increase of dissipation in an ensemble with a fixed coupling force and number of elements can lead to the appearance of chaos as a result of a cascade of period doubling bifurcations of periodic rotational motions or as a result of invariant tori destruction bifurcation. Chaos and hyperchaos can occur in an ensemble by adding or excluding one or more elements. Moreover, chaos arises hard, since in this case the control parameter is discrete. The influence of the coupling strength on the occurrence of chaos is specific. The appearance of chaos occurs with small and intermediate coupling and is caused by the overlap of the various out-of-phase rotational modes regions existence. The boundaries of these areas are determined analytically and confirmed in a numerical experiment. Chaotic regimes in the chain do not exist if the coupling strength is strong enough. The dimension of observed hyperchaotic regime strongly depends on the number of coupled elements.

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Because of many applications, the study of collective dynamics, especially synchronization and chaos are some of central subjects in nonlinear dynamics for the past three decades. Many significant and important results has been obtained. In our paper we present a theoretical and computational study of complex dynamics – chaotic and hyperchaotic rotations – in the chains of locally coupled identical pendulums. The influence of individual (dissipation of pendulums) and collective (coupling strength and number of elements) properties of population is analyzed and discussed.

I. INTRODUCTION

Networks of interacting nonlinear oscillators are encountered in various natural and technical situations. They govern the behavior of coupled neurons and cardiac cells, many physical devices such as arrays of Josephson junctions and lasers, many engineering applications such as phase locked loops and electric power machines. The study of spatio-temporal dynamics in ensembles of nonlinear oscillators of various nature is one of the most popular and interesting directions in modern nonlinear dynamics. The behavior of elements of such ensembles can be roughly divided into three types: (i) fully organized (consistent, coherent, synchronous), (ii) completely disorganized (inconsistent, incoherent, asynchronous), and (iii) intermediate between (i) and (ii) (partially consistent, partially coherent, partially synchronous, such as chimera or cluster states).

Chaotic oscillations are one of the most common phenomena in nonlinear dynamical systems of dimension three and higher. It can be assumed that the main mechanisms of the appearance of chaotic dynamics are currently well understood.[1]. The appearance (and disappearance) of dynamical chaos have been studied in detail for low-dimensional systems[2–6]. From the point of view of bifurcation theory, nothing unusual happens when chaos occurs in large distributed systems – discrete networks and media[7,8]. However, it is clear that the behavior of ensembles not only affects the individual dynamics of its components, but also the different characteristics of the interaction of elements: the type and strength of coupling, configuration and capacity of the network and others.

Hyperchaos is also of interest. A hyperchaotic attractor is typically defined as chaotic behavior with at least two positive Lyapunov exponents[9–13]. Therefore, the dimension of the dynamical systems should be at least 4. In 4-dimensional system the dimension of hyperchaos is equal to 2.

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The phenomenon of hyperchaos is widespread in spatially distributed systems with a large dimension[14–17]. Here appears obvious question: how the dimension of hyperchaos is correlated with the dimension of dynamical system? In our paper we address some comments to this important question.

In this work, the complication of spatio-temporal behavior up to the onset of chaos is investigated in a chain of pendulum-type systems depending on (i) the dissipation of a partial element, (ii) the characteristics of the interaction between the elements – the strength of coupling and (iii) the number of interacting elements. We note that in the limit of very large number of elements chaotic behavior can be considered as spatio-temporal chaos.

The work is structured as follows. Section II describes the studied model of a chain of pendulum elements. In the Section III, in-phase rotational motion and the issue of its stability are considered, asymptotic expressions are given for the boundaries of instability regions depending on the coupling strength of the elements in the chain. Section IV provides a detailed description of possible variants of the in-phase mode unstable regions intersection, depending on the change in the number of elements in the chain. Further, in Section V, a scenario of the development of chaotic dynamics and hyperchaos in a chain with an increase in the dissipation parameter with a fixed number of elements, as well as in the case of an increase in the number of elements with a fixed large dissipation is described. In the Conclusion, the main findings on the presented results of the work are formulated.

II. MODEL

We consider N coupled pendulum-type elements described by a system of ordinary differential equations:

$$\begin{gathered}\ddot{\varphi}_1+\lambda\dot{\varphi}_1+\sin\varphi_1=\gamma+K\sin(\varphi_2-\varphi_1),\\ \ddot{\varphi}_n+\lambda\dot{\varphi}_n+\sin\varphi_n=\gamma+K\left[\sin(\varphi_{n+1}-\varphi_n)+\sin(\varphi_{n-1}-\varphi_n)\right],\ \ n=2,\ldots,N-1,\\ \ddot{\varphi}_N+\lambda\dot{\varphi}_N+\sin\varphi_N=\gamma+K\sin(\varphi_{N-1}-\varphi_N),\end{gathered}\tag{1}$$

where λ – damping parameter, responsible for dissipative processes in the system, γ – constant torque is the same for all N pendulums, K – parameter of coupling strength between elements.

The Eq. (1) is used to describe the behavior of interacting pendulums[18–20], coupled Josephson junctions[21–26]. It is used to describe processes in superconductors[27], molecular biology[28,29], phase

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synchronization systems[30]. Moreover, such a system can be considered as a generalization of the Kuramoto model taking into account the inertia and intrinsic nonlinearity of the ensemble elements[31–34].

Since the elements of the ensemble are identical, the following spatially homogeneous modes exist for any values of the parameters:

- equilibrium state with phase coherent elements $\varphi_1 = \varphi_2 = \ldots = \varphi_N = \overline{\psi} = \text{const}$;
- rotational motion of the elements with coherent phase $\varphi_1(t) = \varphi_2(t) = \ldots = \varphi_N(t) \equiv \psi(t)$.

These modes satisfy the pendulum equation:

$$\ddot{\psi} + \lambda\,\dot{\psi} + \sin\psi = \gamma. \tag{2}$$

Eq. (2) has been carefully examined in[35,36]. In Fig. 1 the parameter plain for Eq. (2) is presented.

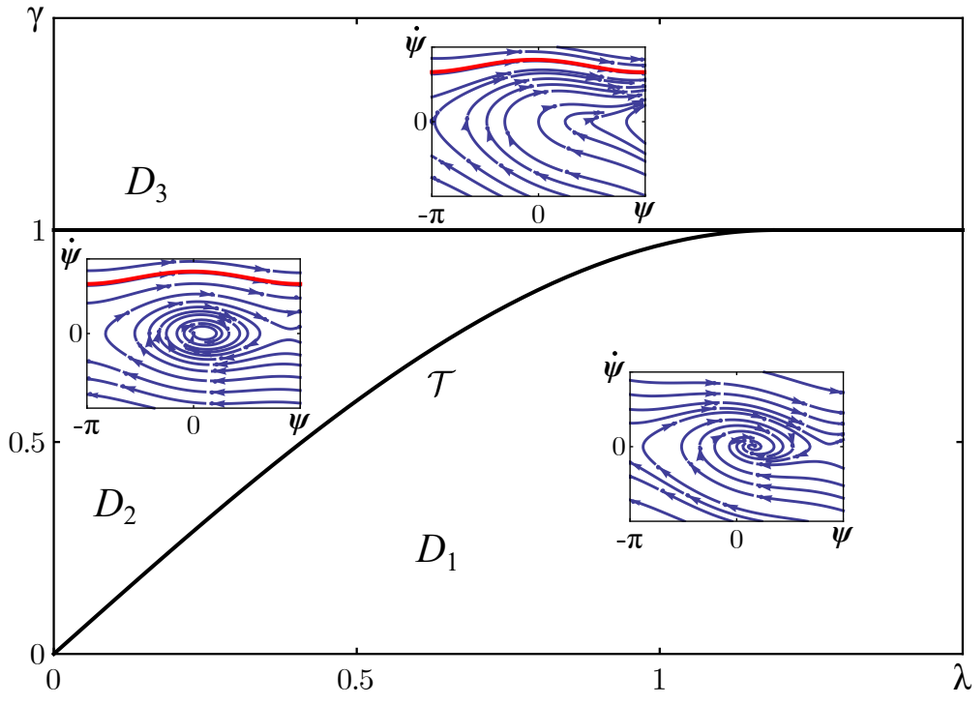

FIG. 1. Bifurcation diagram (plane of parameters λ and γ) and structurally stable phase portraits of the system (2). The $\mathscr{T}$ curve is a Tricomi bifurcation curve, passing through which from D_1 to D_2 the invariant saddle-node curve is destroyed and a rotational 2π-periodic rotation emerges (marked by red lines).

In this study, we are interested in regular and chaotic rotational modes. We focus on the values of the parameters λ and γ, for which in the system (1) there is an in-phase rotation periodic

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motion – in-phase regime (IPR) and stable equilibrium, which corresponds to the area D_2 on the plane (λ, γ), bounded below by the Tricomi bifurcation curve $\mathscr{T}$[35,36].

In our previous works, we considered the features of the rotational dynamics of particular variants of the system (1). In[20,37] two coupled pendulums were investigated for the cases of symmetric and asymmetric coupling, respectively. It was shown that there is a region of instability of the in-phase regime, where out-of-phase rotations arise as a result of the period doubling bifurcation. Further, with an increase of the dissipation parameter in the system chaotic rotations were realized as a result of a cascade of period doubling bifurcations. In the article[38] the case of a three locally coupled elements chain was considered. The existence of two regions of instability of in-phase rotations was shown and the process of approach and intersection of two regions of chaotic dynamics was described in detail. Further in the work[39] a case of N globally coupled pendulums was studied. It was demonstrated that there are $N-1$ regions of instability, the boundaries of which coincide. The phenomena of multistability of out-of-phase rotations, as well as the occurrence of chaos are described. Chaotic rotations arise in the region of parameters where the intervals of instability of the in-phase regime coexist. In this paper, we describe the case of a chain of pendulums with variable length. Here, the regions of instability at low dissipation λ are distant from each other. And hence chaos region will occur at different values of the coupling strength K.

III. IN-PHASE REGIME AND ITS STABILITY

The possibility of implementation of rotational regimes of various complexity degree in system (1) is directly related to the in-phase regime stability issue, the instability development which leads to the out-of-phase regimes (OPR) appearance. For this reason, let us briefly consider the in-phase mode stability issue. For this, we linearize the system (1) in the neighborhood of $\phi(t)$, representing $\varphi_n(t)$ in the form $\varphi_n(t) = \phi(t) + \delta\varphi_n(t)$. The linearized system of equations for perturbations $\delta\varphi_n(t)$ has the form:

$$\begin{gathered}
\delta\ddot{\varphi}_1 + \lambda\,\delta\dot{\varphi}_1 + \cos\phi(t)\,\delta\varphi_1 = K(\delta\varphi_2 - \delta\varphi_1), \\
\delta\ddot{\varphi}_n + \lambda\,\delta\dot{\varphi}_n + \cos\phi(t)\,\delta\varphi_n = K(\delta\varphi_{n-1} - 2\delta\varphi_n + \delta\varphi_{n+1}), \quad n = 2,\ldots,N-1, \\
\delta\ddot{\varphi}_N + \lambda\,\delta\dot{\varphi}_N + \cos\phi(t)\,\delta\varphi_N = K(\delta\varphi_{N-1} - \delta\varphi_N).
\end{gathered} \tag{3}$$

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We pass in the system (3) to the normal coordinates $\psi_1, \psi_2, \ldots, \psi_N$ and obtain the following system of independent equations[39]:

$$\ddot{\psi}_n + \lambda \dot{\psi}_n + [\cos\phi(t) - K\mu_n]\psi_n = 0, \tag{4}$$

where $\mu_n = -2[1+\cos(n\pi/N)]$[40]. In our previous works[20,38,39] it was shown that the modes $\psi_1, \psi_2, \ldots, \psi_{N-1}$ can become unstable, which leads to the appearance of out-of-phase rotational modes, while the boundaries of the instability interval $(K_1^{(n)}, K_2^{(n)})$ are determined by the expressions

$$K_{1,2}^{(n)} = K_{1,2}^{*}/|\mu_n|, \tag{5}$$

where $K_{1,2}^{*}(\lambda,\gamma)$ are, respectively, the left and right boundaries of the instability region of the in-phase solution of the equation $\ddot{\psi} + \lambda\dot{\psi} + [\cos\phi(t) + K^*]\psi = 0$, which can be determined by the asymptotic expressions

$$K_{1,2}^{*} = \frac{1}{4}\left[\frac{\gamma^2}{\lambda^2} \mp 2\sqrt{1-\gamma^2} + \frac{1}{2}\frac{\lambda^2}{\gamma^2}\right] + O\left(\frac{\lambda^4}{\gamma^4}\right). \tag{6}$$

Thus, in the system (1), for certain values of the control parameters λ, γ, an $N-1$ instability region of the in-phase mode $\phi(t)$ can exist. Moreover, these instability regions do not overlap at low dissipation ($\lambda \ll 1$).

IV. INTERSECTIONS OF IN-PHASE REGIME INSTABILITY REGIONS

In this section, we show that the mutual arrangement of the instability regions of the in-phase regime significantly affects the complexity of the dynamical regimes being realized. As a result of direct numerical simulation of the system (1) in the parameter area (λ,γ), maps of rotational modes were built, shown in Fig. 2, analyzing which we see that there is a complication of structures with an increase of dissipation parameter. There is a convergence and overlap of the zones of various structures existence, as well as the emergence of multistability, which ultimately leads to the emergence of a regime of dynamical chaos. Positive largest Lyapunov exponent defines chaoticity of behavior.

Let us analyze how the emergence, evolution and overlapping of the zones of instability of the IPR occurs. We show that for fixed values of the parameters λ and γ increased number of the chain elements N leads to the following scenarios of overlap instability regions:

(i) one of the instability regions is separated from the rest when new elements are added;

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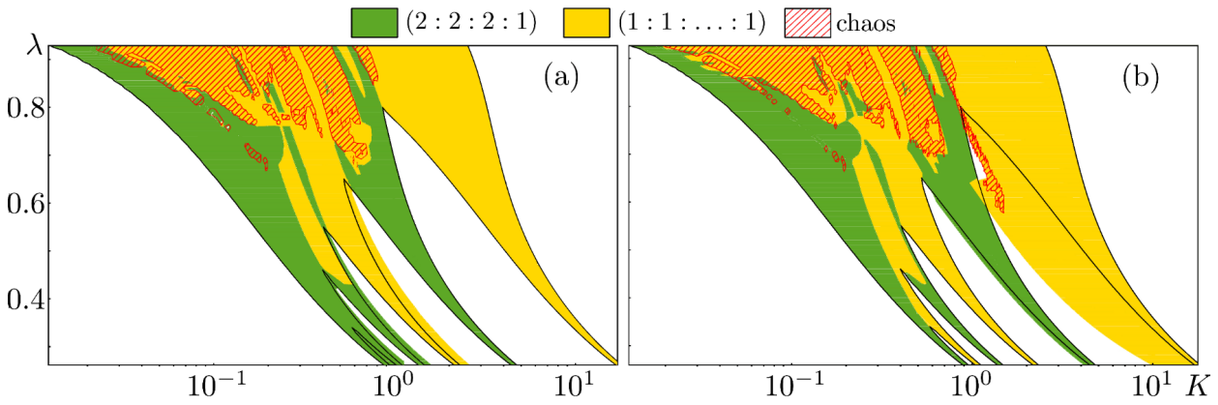

FIG. 2. Maps of rotational modes are realized in the system (1) for $N = 7$, $\gamma = 0.97$ for different values of the λ and K. Black lines – boundaries of instability regions of in-phase rotational motion $\phi(t)$, obtained numerically. The color indicates the types of realized rotational modes: green – $(2:2:2:1)$ regimes, yellow – $(1:1:\ldots:1)$ regimes. Red shading indicates areas of chaos, where the largest Lyapunov exponent takes positive values. Maps are obtained for increasing (a) and decreasing (b) parameter K. In all simulations initial conditions for next parameter values are equal to final states of previous trial. There are rather big areas, where structures of chaotic behavior coexist with fully synchronous rotations.

(ii) there is such a critical number of elements N^*, when an exceeding instability region is separated from the rest;

(iii) for any number of elements N, all regions of instability have intersections with others.

From the expressions (5), (6) and the monotonic increase of the expressions $\mu_1, \mu_2, \ldots, \mu_{N-1}$ it follows that the sequences of the left $K_1^{(1)}, K_1^{(2)}, \ldots, K_1^{(N-1)}$ and right $K_2^{(1)}, K_2^{(2)}, \ldots, K_2^{(N-1)}$ boundaries of instability regions increase monotonically (the width of instability regions $K_2^{(n)} - K_1^{(n)}$ also increases monotonically). Thus, two near regions of instability can partially overlap ($K_1^{(n+1)} \leq K_2^{(n)}$), or be separated by a stability window ($K_1^{(n+1)} > K_2^{(n)}$). Thus, the zone of instability cannot entirely enter into another zone of instability. Let us introduce the number $n^*(N)$, which determines the number of the first overlapping zones of instability with increasing bond strength parameter K. For example, if $n^*(8) = 3$, then the first three zones of instability overlap, then the stability window follows (the location of the other four zones does not interest us, it is important that they are separated from each other). The expression $n^*(8) = 7$ means that all zones of instability have overlapped and merged into one. It's obvious that $1 \leq n^*(N) \leq N-1$. Formally, $n^*(N)$

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is defined by the expression

$$n^*(N) = \min\left(\left\{n \in \{1,2,\ldots,N-2\} \,|K_1^{(n+1)} - K_2^{(n)} > 0\right\} \cup \{N-1\}\right). \tag{7}$$

To find an explicit expression for the number $n^*(N)$, consider the expression continualized by n: $K_1^{(n+1)} - K_2^{(n)}$, introduce the distance function between instability regions $L(x) = K_1^{(x+1)} - K_2^{(x)}$. Obviously, the function $L(x)$ is periodic: $L(x+2N) = L(x)$. In addition, there are gaps at the points $x = (2k+1)N$ and $x = (2k+1)N - 1$, where k is integer. It can be verified that $\lim\limits_{x\to N} L(x) = -\infty$ and $\lim\limits_{x\to N-1} L(x) = +\infty$. Thus, the intervals $[-N, N-1]$ and $[N-1, N]$ contain at least one root of the equation $L(x) = 0$. The direct solution of the equation $L(x) = 0$ leads to the roots

$$x_k^{\pm} = \frac{2N}{\pi}\arctan\left(\frac{\cos\left(\frac{\pi}{2N}\right) \pm \sqrt{K_1^*/K_2^*}}{\sin\left(\frac{\pi}{2N}\right)}\right) + 2Nk. \tag{8}$$

The roots x_0^- and x_0^+ are on the interval $[-N, N]$. We conclude that each interval $[-N, N-1]$ and $[N-1, N]$ contains one root, and the smaller root x_0^- lies in the interval $[-N, N-1]$. Thus, for $x > x_0^-$ we have $L(x) > 0$: the length of the gap between the regions of instability becomes positive, those zones no longer overlap. Then

$$n^*(N) = \max\left\{1, \quad \left\lceil \frac{2N}{\pi}\arctan\left(\frac{\cos\left(\frac{\pi}{2N}\right) - \sqrt{K_1^*/K_2^*}}{\sin\left(\frac{\pi}{2N}\right)}\right)\right\rceil\right\}, \tag{9}$$

where the up-square brackets $\lceil\ldots\rceil$ mean ceiling function, that rounds the value to the nearest integer which is not less than the given value. Thus, as the parameter K increases, only the first $n^*(N)$ instability regions can overlap. All other unstable regions (if they exist) are separated from each other by stability windows. It is also useful to note that due to the monotonic growth of the function under the ceiling function brackets in Eq. (9), as the number N grows, the sequence $n^*(N)$ is non-decreasing. In other words, an increase of the chain number of elements N can only lead to an increase in the number of intersecting regions of instability of IPR.

Let us define the right boundary $K_2^{(n^*)}$ of the instability zone formed by the overlapping of the instability areas:

$$K_2^{(n^*)} = \frac{K_2^*}{4\cos^2\left(\frac{n^*(N)}{N}\frac{\pi}{2}\right)}. \tag{10}$$

When $N \to \infty$, there is a limit $\lim_{N\to\infty} n^*(N)/N = 1$. In fact, for sufficiently large N (when $\cos\left(\frac{\pi}{2N}\right) > \sqrt{K_1^*/K_2^*}$)

$$n^*(N) = \left\lceil \frac{2N}{\pi}\arctan\left(\frac{\cos\left(\frac{\pi}{2N}\right) - \sqrt{K_1^*/K_2^*}}{\sin\left(\frac{\pi}{2N}\right)}\right)\right\rceil, \tag{11}$$

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the following inequalities are satisfied

$$\frac{2}{\pi}\arctan\left(\frac{\cos\left(\frac{\pi}{2N}\right)-\sqrt{K_1^*/K_2^*}}{\sin\left(\frac{\pi}{2N}\right)}\right)\leq\frac{n^*(N)}{N}\leq 1-\frac{1}{N}. \tag{12}$$

Then, calculating the limits on both sides, by the two attendant theorem we arrive at the indicated limit. As a result, we arrive at

$$\lim_{N\to\infty}K_2^{(n^*)}=+\infty. \tag{13}$$

The left boundary of the instability zone formed by the overlap of individual instability regions is defined as $K_1^{(1)}$. When $N\to\infty$,

$$\lim_{N\to\infty}K_1^{(1)}=K_1^*/4. \tag{14}$$

For a more detailed study of the behavior of the instability regions we consider the number $\overline{n}^*(N)=N-1-n^*(N)$ $(0\leq\overline{n}^*(N)\leq N-2)$, that determines the number of isolated (by stability windows) unstable regions. From the expression (9) we get that

$$\overline{n}^*(N)=\min\left\{N-1,\quad\left\lfloor\frac{2N}{\pi}\left(\frac{\pi}{2}-\arctan\left(\frac{\cos\left(\frac{\pi}{2N}\right)-\sqrt{K_1^*/K_2^*}}{\sin\left(\frac{\pi}{2N}\right)}\right)\right)\right\rfloor\right\}-1, \tag{15}$$

where the down-square brackets $\lfloor\ldots\rfloor$ mean floor function, that rounds the value to the nearest integer which is not greater than the given value. Let's examine the behavior of the function under the floor function brackets. Direct computation can verify that its second derivative with respect to N is always negative for $\forall N\geq 1$ (we are interested in the values $N\geq 2$). Thus, for $N\geq 1$, the first derivative decreases monotonically with increasing N. At the point $N=1$, the derivative is $\frac{K_1^*}{K_1^*+K_2^*}+\frac{2}{\pi}\arctan\sqrt{\frac{K_1^*}{K_2^*}}>0$; as $N\to+\infty$, the value of the derivative tends to $+0$. Therefore it can be argued that the derivative all $N\geq 1$ is positive and of the function increases monotonically with an increase of the number $N\geq 2$. Then the sequence $\overline{n}^*(N)$, like the sequence $n^*(N)$, is non-decreasing.

Scenario A. Let us find a condition under which an isolated region of instability of IPR exists for any N. From the non-decreasing sequence $\overline{n}^*(N)$ it follows that a necessary and sufficient condition is the fulfillment of the equality $\overline{n}^*(3)=1$. From Eq. (15) we find $K_2^*/K_1^*<3$. The example of this scenario is demonstrated in Fig. 3(a).

Scenario B. Similarly, one can find a condition under which all regions of instability overlap for any N. A necessary and sufficient condition is the equality $\lim_{N\to+\infty}\overline{n}^*(N)=0$. From Eq. (15) it follows $\lim_{N\to+\infty}\overline{n}^*(N)=\left\lfloor\left(1-\sqrt{K_1^*/K_2^*}\right)^{-1}\right\rfloor-1$. Then we get $K_2^*/K_1^*>4$ (see Fig. 3(b)).

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Scenario C. When $3 < K_2^*/K_1^* < 4$ for small N all instability regions overlap, but starting from some number N isolated regions appear separated by stability windows (see Fig. 3(c)).

Another conclusion from the non-decreasing sequences $n^*(N)$ and $\overline{n}^*(N)$ is the fact that when

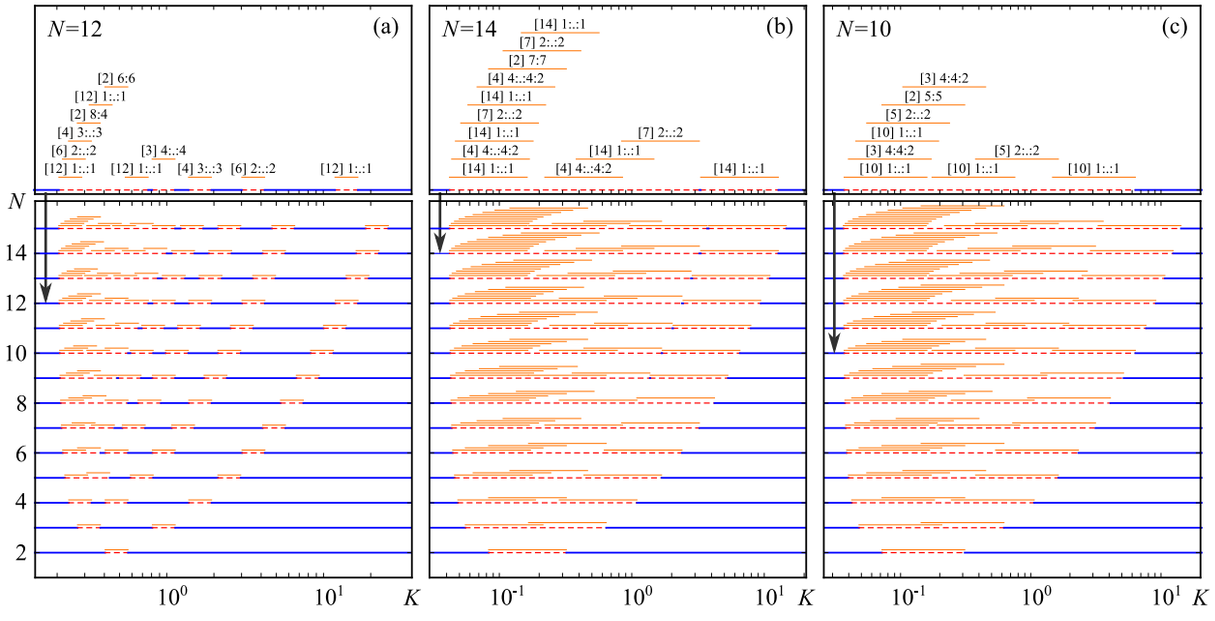

FIG. 3. Lower panels. The regions of instability of the ψ_n modes (orange solid lines), and the corresponding regions of instability of the IPR (red dashed lines), found from the expression (5) and the numerical calculation of the boundaries $K_{1,2}^*$, for $\gamma = 0.97$ and different N and λ depending on the coupling strength K. Blue solid lines represent stable IPR. Top panels. Instability regions of ψ_n modes and the corresponding types of OPR for indicated by arrow number N. The number in square brackets indicates the number of in-phase clusters. (a) $\lambda = 0.5$. Scenario A. For any N, there is at least one region of in-phase instability, isolated from the others by stability windows. (b) $\lambda = 0.8$. Scenario B. Since some N (here $N = 6$) appears common mode instability region isolated from other stability windows. (c) $\lambda = 0.82$. Scenario C. For any number of elements N, overlap of all instability regions of the ψ_n modes is observed with the formation of instability area of the in-phase mode without stable windows. Over panel. Instability regions of ψ_n modes and the corresponding types of out-of-phase rotation at $N = 15$. The number in square brackets indicates the number of in-phase clusters.

adding a new element, i.e. as N increases by one, either one new isolated region of instability appears and the number of overlapping regions does not change, or the number of overlapping regions increases by one, but the number of isolated regions does not change.

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V. THE CHAOTIC DYNAMICS DEVELOPMENT

Here we analyze in detail the complex periodic and chaotic regimes occurrence mechanisms, and the various OPRs existence regions overlapping role. Below we present the results of computations. First, we introduce the synchronicity parameter

$$\Xi = \frac{1}{N(N-1)} \sum_{n_1,n_2=1}^{N} \max_{0\le t\le T} \left|\dot{\varphi}_{n_1}(t) - \dot{\varphi}_{n_2}(t)\right|, \tag{16}$$

which characterizes the rotational motion phase synchronization degree. The value $\Xi = 0$ shows that the rotational regime under consideration is in-phase, the values $\Xi > 0$ indicate the realization of OPR. The types of rotational modes arising in the system will be denoted as $(a_1 : a_2 : \ldots : a_m : \ldots : a_M)$, where the number a_m shows the number of elements with the same phase in the m-th cluster. The in-phase mode in a chain of N elements is denoted by $(N : 0)$. For a more detailed analysis, we present bifurcation diagrams, as well as graphs that show the local maxima of the oscillator frequencies and the largest Lyapunov exponent, which positive value indicates the dynamical chaos regime presence in the system.

Let us investigate the rotational regimes dynamics in the chain depending on the parameter K for different dissipation parameter λ values and different number of elements N. Our computational experiments show that with a parameter K value increase the IPR loses its stability softly, but the corresponding OPR stability loss with a further increase in K occurs in a hard manner. Thus, hysteresis occurs when there is a range of K values where in-phase mode coexists with out-of-phase mode. We call such an interval of in-phase regime instability as "right". If the in-phase mode loses its stability hardly and the out-of-phase mode loses its stability softly with K increasing then we call such an instability range "left". The scenario for the appearance and existence of "left" and "right" regions for a different number of elements N is as follows:

$N = 2$ – one "left" $(1 : 1)$ region;

$N = 3$ – one "left" $(2 : 1)$, one "right" $(1 : 1 : 1)$;

$N = 4$ – two "left" $(1 : 1 : 1 : 1)$ and $(2 : 2)$, one "right" $(1 : 1 : 1 : 1)$;

$N = 5$ – two "left" $(2 : 2 : 1)$ and $(1 : 1 : 1 : 1 : 1)$, two "right" $(2 : 2 : 1)$ and $(1 : 1 : 1 : 1 : 1)$; etc.

Thus, the transition from an even number of elements to an odd number is accompanied by the addition of one "right" area; similarly, from an odd number to an even number one "left" area is added.

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A. Fixed number of elements N

Let us consider the chain dynamics for $N = 7$, $\gamma = 0.97$, $\lambda = 0.3$ (see Fig. 4). In this case, with the parameter K value increase the out-of-phase $(2:2:2:1)$ and $(1:1:\ldots:1)$ regimes associated with the IPRs instability development and corresponding to the ψ_n $(n = 1,\ldots,6)$ modes are sequentially realized in the system (1). Note that modes with the same designation can differ from each other and represent different objects in a phase space of the system (1). Let us consider the range of the parameter K values when the rotational motion $(2:2:2:1)$ is realized for the first time. Here, as the parameter K increases, the in-phase periodic rotation $\phi(t)$ undergoes a period-doubling bifurcation at $K \approx 0.65689$. In this case, stable in-phase 2π-periodic motion gives rise to stable 4π-periodic $(2:2:2:1)$ motion, but $\phi(t)$ loses its stability. The bifurcation diagram (see Fig. 4(a)) shows that there is also an unstable motion $(2:2:2:1)$, which at $K \approx 0.72420$ arises from an unstable in-phase motion as the period doubling bifurcation result, while $\phi(t)$ becomes stable again. Further, as the parameter K increases, stable and unstable $(2:2:2:1)$ motions merge and disappear as a result of saddle-node bifurcation. With a further increase in the coupling parameter K the regimes $(1:1:\ldots:1)$ and $(2:2:2:1)$ appear in a soft manner, and then regimes $(1:1:\ldots:1)$, $(2:2:2:1)$, $(1:1:\ldots:1)$ appear in a hard manner. Thus, when the parameter K changes there are three "right" and three "left" in-phase regime $\phi(t)$ instability regions in the system (1) with small dissipation λ. Note also that for $\lambda = 0.3$ 4π-periodic rotational regimes $(2:2:2:1)$ and $(1:1:\ldots:1)$, referring to "right" instability intervals, can coexist on the interval of the coupling strength $0.76917 \leq K \leq 0.90117$, although the corresponding instability regions do not intersect. In considered case the appearance not only different periodic rotation modes is possible. In interval $1.95 < K < 2.0$ we observe stable invariant torus that occurs due to Neimark-Sacker bifurcation.

A further increase of the dissipation parameter λ leads to the in-phase mode instability intervals intersections appearance. For $\lambda = 0.6$ four regions corresponding to the smallest coupling parameter K values intersect (see Fig. 5), while the 4π-periodic motions corresponding to the $(1:1:\ldots:1)$ regime are unstable and rotations with a larger number of turns by 2π, which arise during subsequent 4π-periodic trajectories bifurcations, are realized. In the "left" region corresponding to $(1:1:\ldots:1)$ regime with chaotic motions appears as a result of a period doubling bifurcations cascade, what is confirmed by the positive values of the largest Lyapunov exponent (see Fig. 5(c)). In the case $\lambda = 0.7$ (see Fig. 6), when five instability regions intersect, chaotic

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regimes are observed already at several intervals. The transition from periodic to chaotic dynamics can occur through the Neimark-Sacker bifurcation leading to the emergence of an invariant torus, and its subsequent destruction through the torus destruction bifurcation which ultimately leads to the chaotic attractor birth. Here, the curves corresponding to 4π-periodic rotational regimes have a rather nontrivial structure with different closures with each other (see Fig. 6(a)). With a further increase in the dissipation parameter λ the regions of chaotic dynamics gradually increase and merge with one another (see Fig. 7(b)).

B. Fixed dissipation parameter λ

Let us consider the regimes evolution in the case of the fixed dissipation parameter value, for example, $\lambda = 0.9$, with a change in the number of elements N and the coupling parameter K. First of all, we are interested in the transitions from regular to chaotic regimes and the role played by the appearance and evolution of the IPR instability regions during these transitions.

Using the general expression (5), we can determine the number N_ψ and the fraction $u_N(K) = N_\psi/(N-1)$ of modes ψ_n that are unstable for some fixed value of the coupling parameter K:

$$u_N(K) = \frac{\theta(K-K_1^*/4)}{N-1}\left\lfloor \frac{2N}{\pi}\arccos\left(\sqrt{\frac{K_1^*}{4K}}\right)\right\rfloor - \frac{\theta(K-K_2^*/4)}{N-1}\left\lfloor \frac{2N}{\pi}\arccos\left(\sqrt{\frac{K_2^*}{4K}}\right)\right\rfloor, \qquad (17)$$

where θ is the Heaviside step function. Since $K_2^*/K_1^* > 4$, then for any N all instability regions of ψ_n modes overlap, which leads to the formation of a single, global IPRs instability region. Find out what happens in the limit $N \to \infty$. For this case, we introduce the fraction $u_\infty(K)$ determined by the expression

$$u_\infty(K) = \frac{2}{\pi}\left[\theta(K-K_1^*/4)\arccos\left(\sqrt{\frac{K_1^*}{4K}}\right) - \theta(K-K_2^*/4)\arccos\left(\sqrt{\frac{K_2^*}{4K}}\right)\right]. \qquad (18)$$

Fig. 7(c) demonstrates the dependence of the unstable modes fraction on the coupling parameter K for fixed $N = 7$ and in the limit $N \to \infty$. The figure shows that the largest unstable modes number is observed at small and intermediate coupling parameter values ($0.01 < K < 1$). As a result of the IPR stability loss new OPR appear, which in turn can also become unstable, which leads to the new and new OPRs emergence (see Figs. 4–7).

The general property of the obtained dependence (18) is its maximum at the point $K = K_2^*/4$, which is demonstrated in the Fig. 7(c), obtained for the considered case $\lambda = 0.9$. Thus, the maximum ψ_N mode instability regions overlapping density is observed at small and intermediate K

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values located near the left boundary of the in-phase rotation instability interval, i.e. where the "left" instability regions appear.

Our computational experiments show that chaotic regimes precisely observed in the area of greatest accumulation and intersection of various instability regions. The Fig. 7(a) shows the area where chaotic dynamics is observed on the (K,N) plane at $\lambda = 0.9$. It can be seen that although the right boundary $K_2^{(N-1)}$ of the instability region moves rather quickly to the right as N increases, chaotic regimes are realized only in a rather narrow range of coupling strength values $0.01 < K < 1$. At small K, due to weak interaction, chaotization of initially regular rotations does not occur. In the case of strong couplings, the pendulums interaction leads to the rotations regularization. In this case, the resulting modes are not necessarily in-phase. The in-phase rotational mode is established only with a large coupling strength (see the $K_2^{(N-1)}$ curve in Fig. 7(a)).

Analyzing Fig. 7, the following conclusions can be drawn. For a fixed value of K with a change in the chain length N, two main scenarios of the chaotic behavior emergence and existence are realized:

(i) for small $K \sim 0.1$, the chaos appearing due to the period doubling bifurcation at $N = 2$ does not disappear with an increase in N. The complication of the realized modes occurs with an increase of the number of interacting pendulums. In this case there is an increase of the dimension of chaotic oscillations. Dimension of hyperchaos, i.e. the number of positive Lyapunov exponents, can be approximated by a linear function (see Fig. 8). Note that adding one pendulum to the chain does not always lead to the appearance of a new positive Lyapunov exponent.

(ii) for $0.12 \lesssim K \lesssim 1.0$: a) the addition of one (sometimes two or more) new elements can lead to both chaos and regularization of modes; b) when two (sometimes more) elements are added, the chaotic regime leaves it chaotic, and the regular regime remains regular. The number of elements parity plays an important role here. In this case, the closer K is to 1, the less dense of the overall chaos area becomes, i.e. the regular behavior islands number and size increase. For $K > 1$, no chaotic behavior was found for any N.

A series of computational experiments was carried out in which the chain length N varied with time. In one series, the number of elements was changed by adding (or excluding) one element. In the second series, several elements were added to the chain (excluded from the chain). In all cases, both of the above scenarios were observed.

For fixed N, the regular and chaotic regimes evolution with increasing K is qualitatively the same for different chain lengths N. Namely, chaos appears practically at the same value of $K \approx$

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0.01 and is realized continuously without regular behavior windows up to $K \approx 0.12$. Further, the chaotic regime can alternate with the regular one.

In studied chain of pendulums there is the opportunity of existence of hyperchaos. We computed the spectrum of Lyapunov exponents in the chain of length $N = 7$ and $\gamma = 0.97$, $\lambda = 0.9$ in dependence on coupling strength K (see Fig. 7(c)). There are several intervals of K there four, three, two and one Lyapunov exponents are positive. The areas of existence of stable hyperchaotic regimes is strongly correlated with the areas of the largest number of unstable modes ψ_n. Indeed, as it can be seen in the Fig. 7c, the intervals of values of the coupling parameter K, in which the number of intersections of instability regions N_ψ reaches the highest values (the highest values of u_7), correspond to the largest number of positive Lyapunov exponents N_Λ.

Fig. 8 shows in more detail the relationship between the number of positive Lyapunov exponents N_λ and the number of in-phase instability regions N_ψ depending on the number of elements N. It is seen that for the case of large density values u_N (Fig. 8a,b) the number N_Λ coincides with N_ψ for practically all N and increases linearly with increasing N. For lower values of density u_N (Fig. 8c) the number N_Λ generally does not exceed N_ψ, but both dependencies retain a linear growth trend. Presented results clearly show that the dimension of hyperchaotic regime strongly depends on the number of instabilities of different periodic rotations: each instability transition of periodic behavior added new positive Lyapunov exponent.

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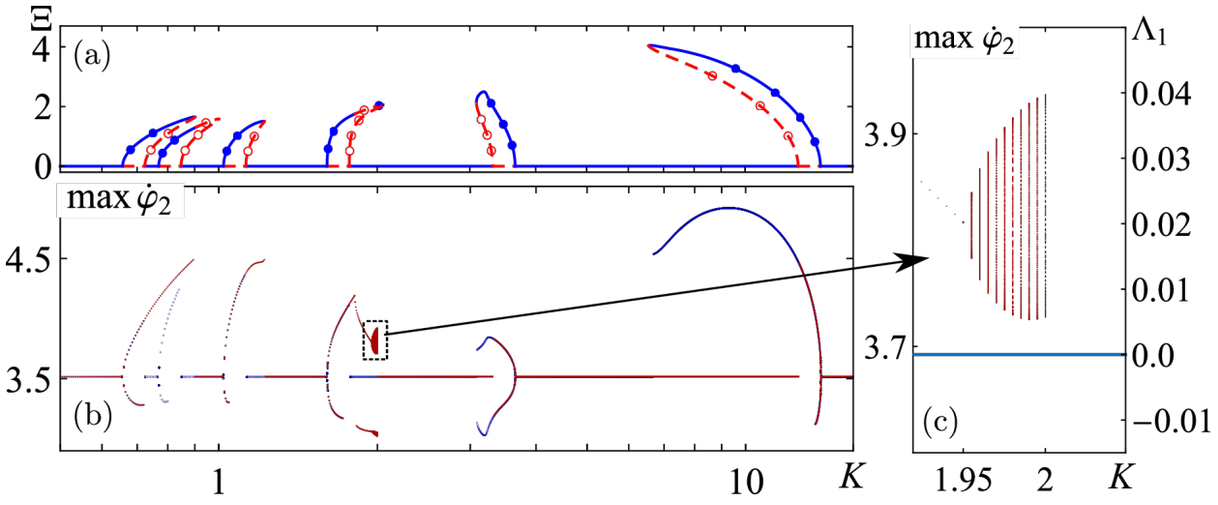

FIG. 4. Local bifurcation diagram of periodic rotational regimes (a). Ξ – synchronicity parameter. Round markers show 4π-periodic rotational regimes. Filled markers correspond to stable rotational regimes, unfilled markers – to unstable ones. The line without markers corresponds to the in-phase 2π-periodic rotational regime, the solid line, to the stable one, and the dotted line, to the unstable one. Local frequency maxima $\max\dot{\varphi}_m$ (b). Diagram is obtained in two ways: with increasing (red markers) and decreasing (blue markers) parameter K. In (c) the enlargement of some area of (b) and the largest Lyapunov exponent $\Lambda_1 = 0$ (blue line) are presented. Fragment (c) shows the emergence of an invariant torus from an out-of-phase 4π-periodic rotational regime. Parameters: $N = 7$, $\gamma = 0.97$, $\lambda = 0.3$, $m = 2$.

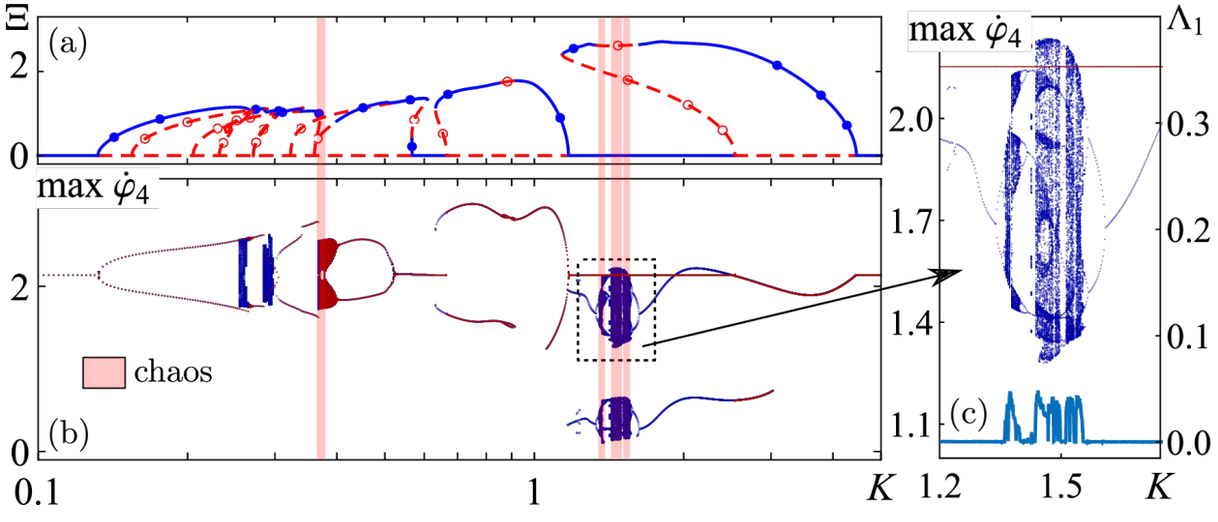

FIG. 5. The same as in Fig. 4. Chaos area is depicted by red color. Fragment (c) shows the emergence of a chaotic state from an out-of-phase 4π-periodic rotational regime as a result of a period doubling bifurcations cascade. Parameters: $N = 7$, $\gamma = 0.97$, $\lambda = 0.6$, $m = 4$.

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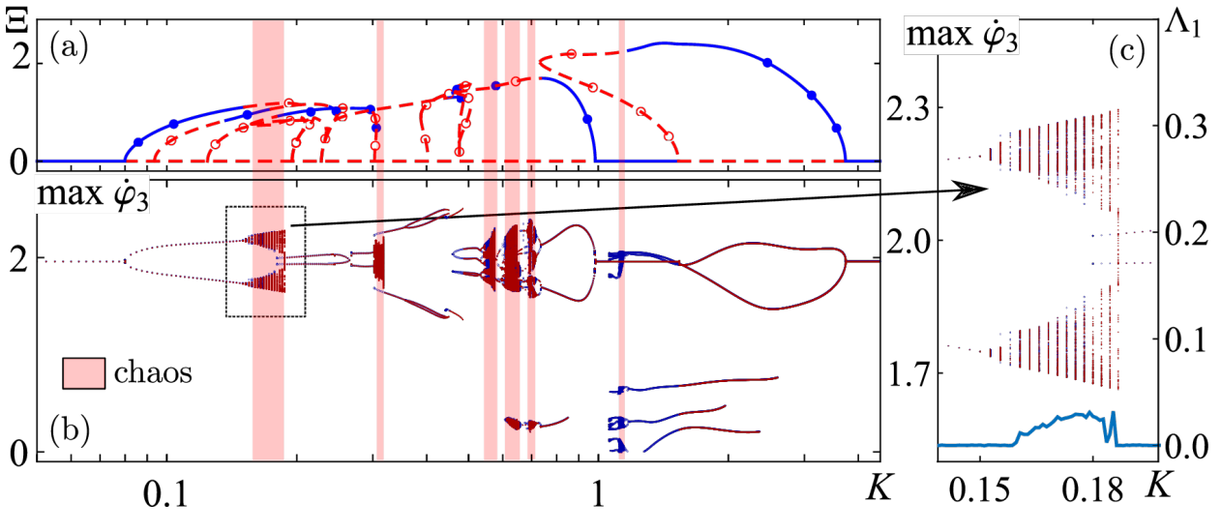

FIG. 6. The same as in Fig. 5. Fragment (c) shows the emergence of a chaotic state from an invariant torus as a result of its destruction. Parameters: $N = 7$, $\gamma = 0.97$, $\lambda = 0.7$, $m = 3$.

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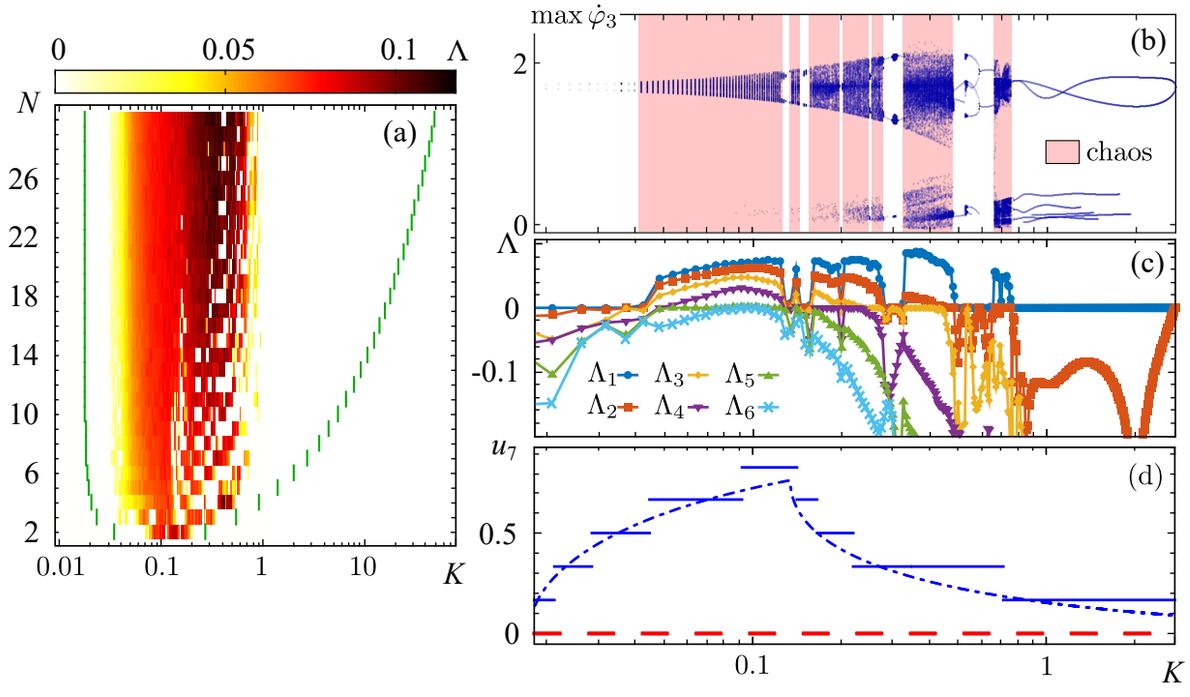

FIG. 7. (a) Chaotic regimes map in the system (1) depending on the coupling strength K and the number of elements N at $\gamma = 0.97$ and $\lambda = 0.9$. Green lines denote the left $K_1^{(1)}$ and right $K_2^{(N-1)}$ boundaries of the IPR $\phi(t)$ instability range. White color denotes regular rotational modes (largest Lyapunov exponent $\Lambda = 0$). In colored area the largest Lyapunov exponent is positive. (b) Local frequency maxima max $\dot{\varphi}_m$ for $m = 3$, $N = 7$. (c) The spectrum (first six maximal) of Lyapunov exponents Λ_j for $N = 7$. There are intervals with four, three, two and one positive exponents. (d) Dependence of fraction $u_7(K)$ of unstable modes ψ_n for $N = 7$ (blue lines) and for $N \to \infty$ (blue dashed curve). The red dashed line indicates the in-phase regime instability region for $N = 7$.

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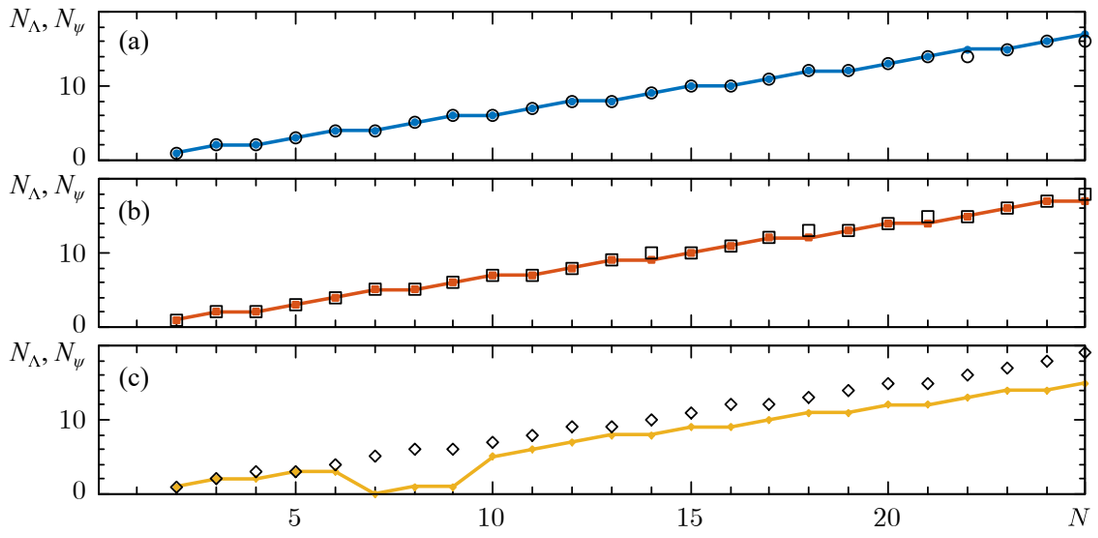

FIG. 8. The quantity of positive Lyapunov exponents N_Λ (colored lines with filled markers) and the number N_ψ of unstable modes ψ_n (unfilled black markers) as the functions of the chain length N for different coupling stength K values: (a) $K = 0.07$, (b) $K = 0.1$, (c) $K = 0.13$. Other system parameters: $\lambda = 0.9$, $\gamma = 0.97$.

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VI. CONCLUSION

In this work we study the features of the appearance of chaotic rotational regimes, including hyperchaotic ones, in a chain of coupled identical pendulums with a variable number of elements N. It is shown that the emergence of these rotations is associated with the development of the instability of the in-phase mode, the number of which is $N-1$. The appearance of out-of-phase regular and chaotic regimes occurs in the region of parameters, where these areas are close to each other or overlap. Analytical estimates of the number of areas with intersections were obtained.

The appearance of chaos occurs as a result of a cascade of period doubling bifurcations or torus destruction bifurcations. Scenarios of the development of chaos are considered in the framework of the changes (a) the dissipation parameter λ and (b) the number of chain elements N. For small values of the dissipation parameter, chaotic rotations begin to appear in non-intersecting regions of instability of the in-phase regime as they approach each other. Then these areas gradually approach and begin to intersect until they merge into one. The zones of chaotic rotations behave similarly, which eventually merge into one (excluding stability windows). This is because with an increase of the dissipation parameter λ (this leads to an approach to the Tricomi curve), the individual pendulum rotations become significantly inhomogeneous in time.

Chaos is observed in the system starting from two elements in the case of strong dissipation. Further, a change in the number of pendulums can lead to both the regularization of the rotational regime and its complication, i.e. to an increase in the number of positive Lyapunov exponents (hyperchaos mode). It was found that their number is directly linearly proportional to the number of overlapping regions of in-phase instability. Instability of some rotational mode added new positive Lyapunov exponent. We especially note that, regardless of the number of pendulums N in the chain, the region of existence of chaotic rotations is localized in a limited range of values of the coupling force K. Outside this region, only regular in-phase and out-of-phase rotations are realized.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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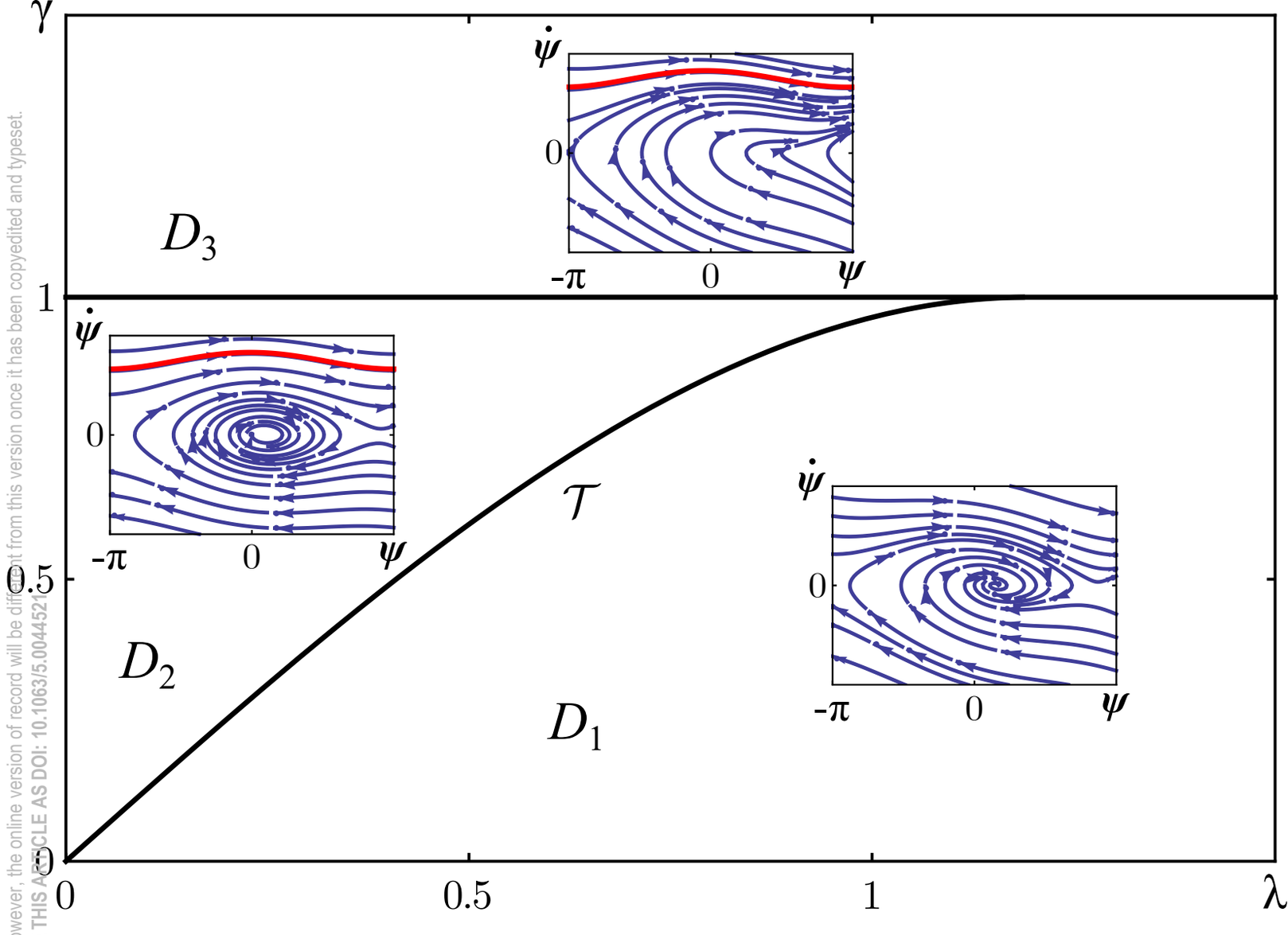

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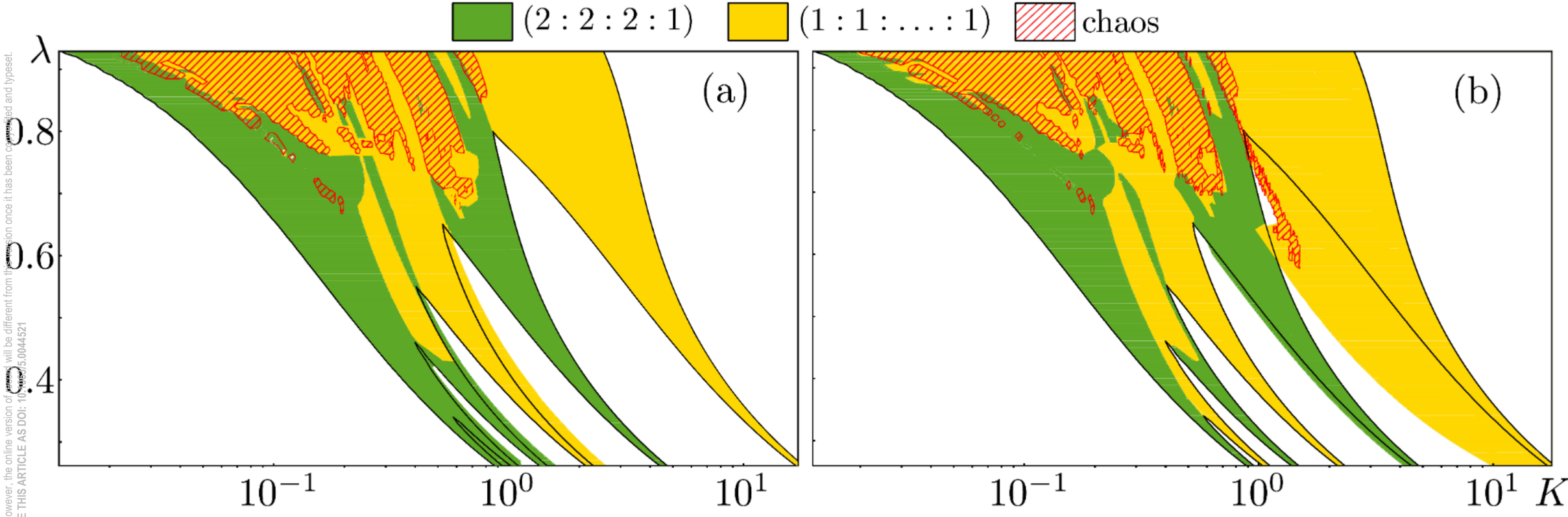

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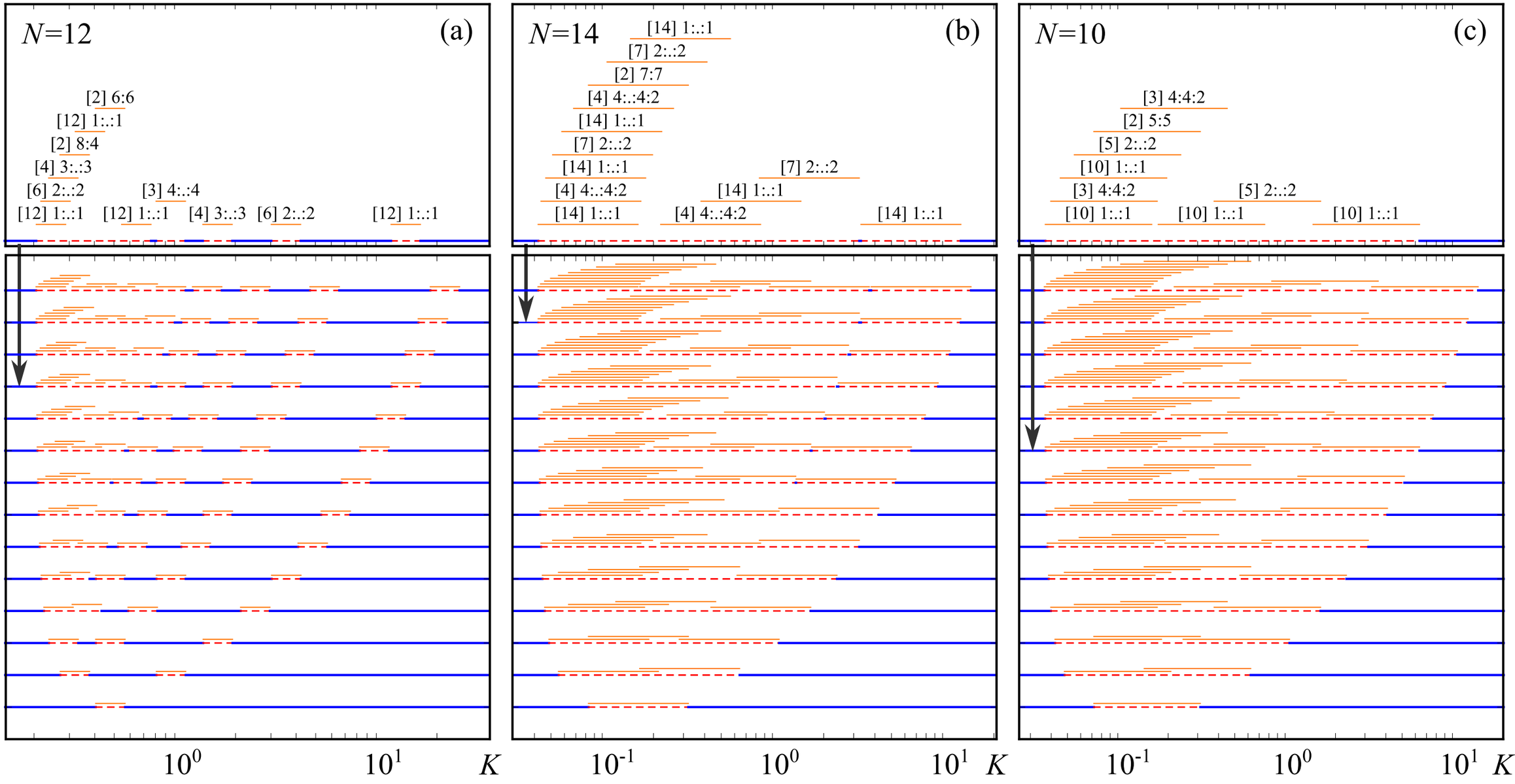

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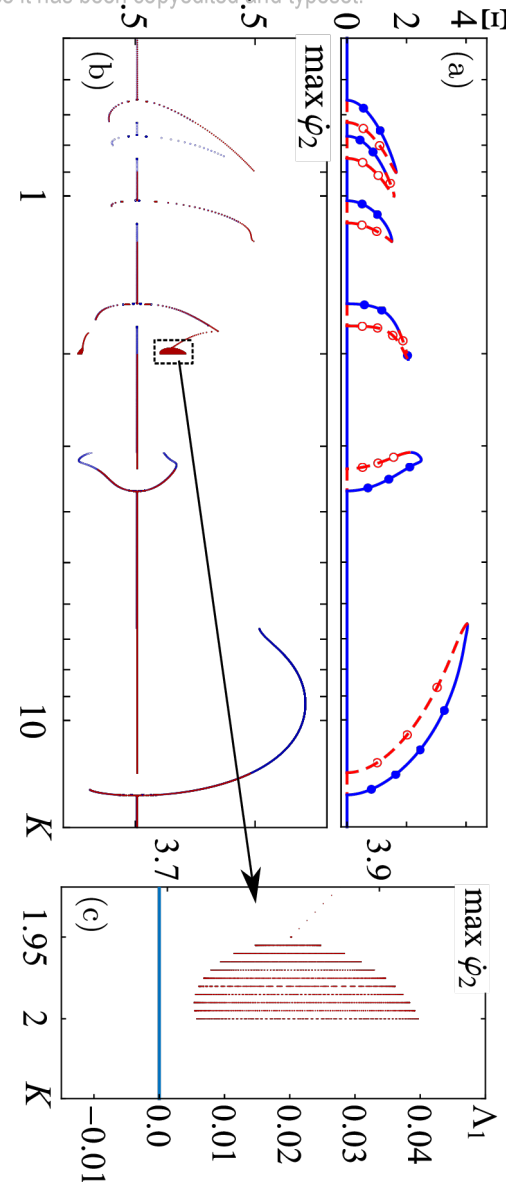

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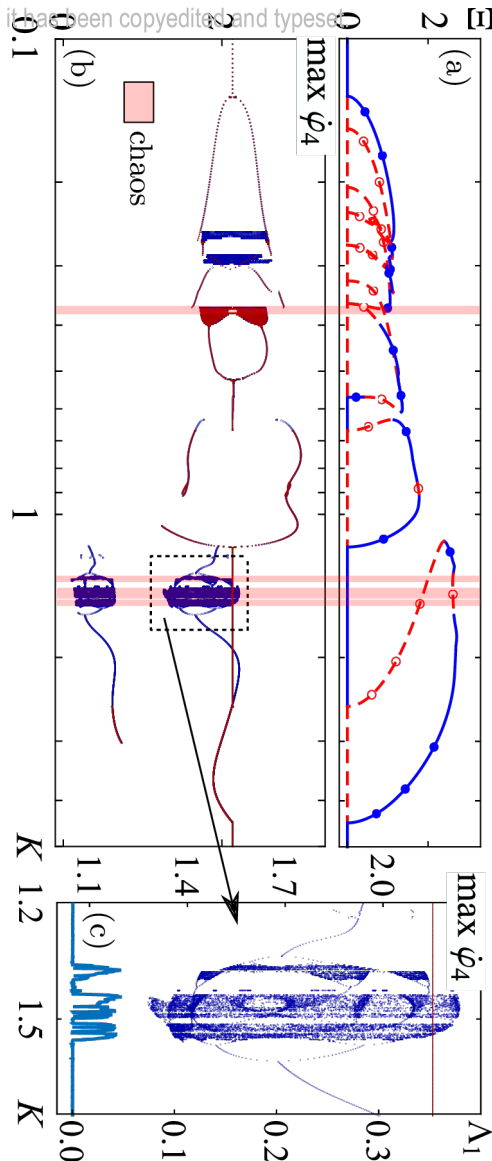

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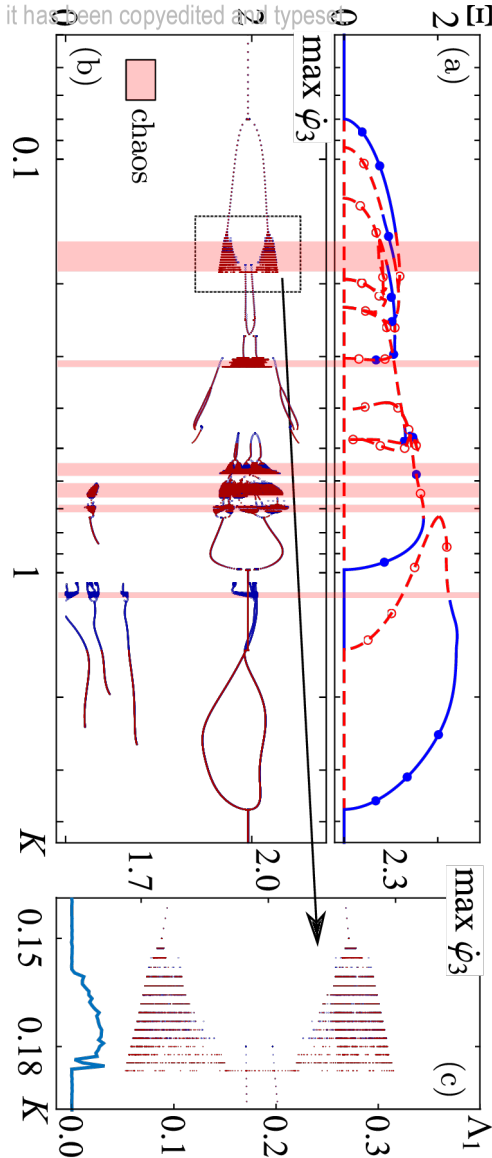

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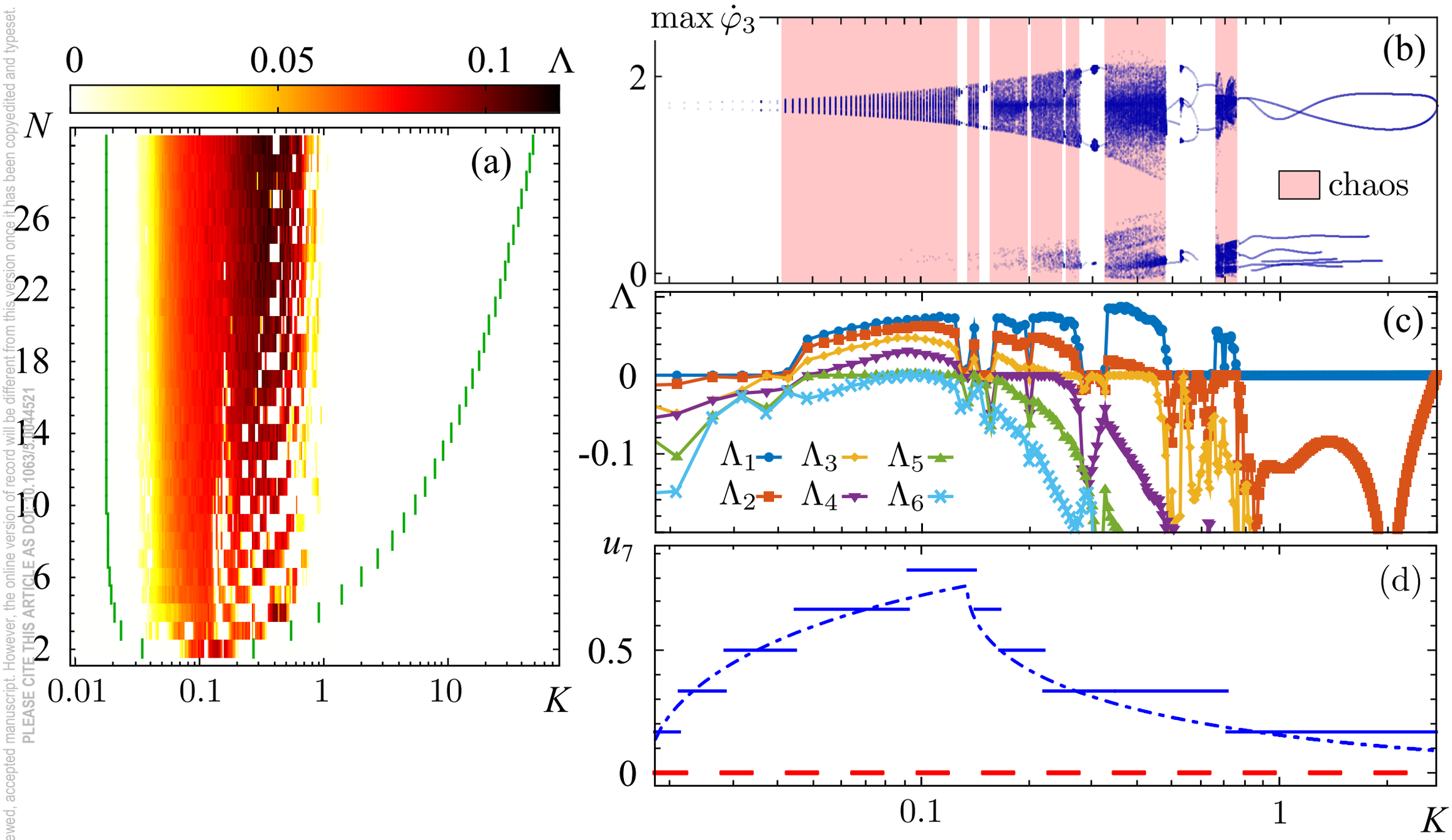

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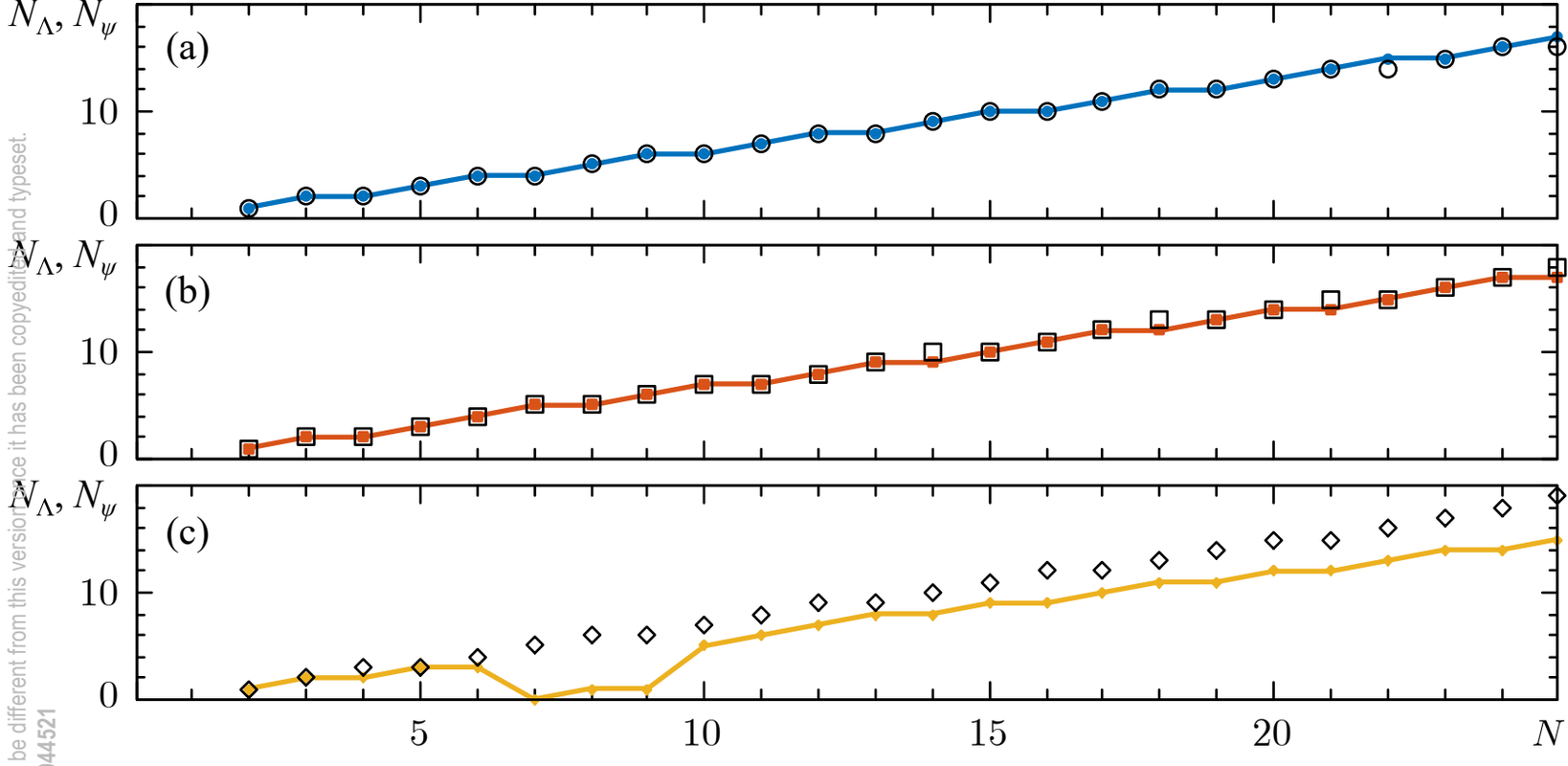

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